



Backlog Parts Staging Containers For HMS

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The backlog management process starts when the recorded defect gets converted into a work order, put into a work plan, and parts are then ordered for the execution of the job. If the ordered parts are not available on site, they will have to be backordered from the Dealer main warehouse or from a Caterpillar distribution center. Backordered parts can take several days to arrive, depending on availability and source; parts may also be non-Caterpillar outside purchases. As the ordered parts arrive at different times, they are stored in the backlog parts staging area. When all the parts for the job have arrived, the job will have a “Parts Order Complete” stamped either manually or electronically and the job can then be scheduled for the work to be completed, generally when the machine is down for a PM Service. In most instances, there will be more than one backlog job.

In most instances, the parts are placed on shelves or in boxes with the work orders. Some mine sites will have a box for each machine and identified by plant numbers or serial numbers. Many cases, the boxes used are too small even for a minor PM Service. This Best Practice focuses specifically on hydraulic mining shovels as it is common to place parts in secondary locations due to their size – Introducing the risk for these to get misplaced or missed when the machine is ready for service work.

2.0 Best Practice Description

This Best Practice describes the use of large portable containers for backlog job parts. These containers are typically 600 to 1000 liters in capacity (The ones shown in this document are approximately 780 liters.) Dividers can be used when there are multiple jobs for a machine. This will help the technicians to easily sort out the required parts for each job. The containers can even be purchased with lids for contamination control or with castor wheels for manual handling instead of utilizing a fork lift truck.

The containers will be able to be moved with a manual pallet fork or powered fork lift truck. If manual handling is required, they should be fitted with castor wheels. They can then be easily slid out of a row for stocking the parts inside. When the scheduled machine is ready to be moved into the workshop for the jobs to be done, the whole container can be moved to the bay and placed next to the machine for unloading the parts.

All backlog jobs are planned, and Maintenance Department will therefore have time to organize the parts to be in the bay before the machine arrives. The size of these containers should therefore not be an issue in transferring from the warehouse to the workshop bay. In fact, the use of these containers re-enforces the use of a single trip to the workshop bay.

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Fig. 1: Typical backlog parts containers stacked up



Fig. 2: Typical backlog parts containers at ground level



Fig. 3: Use of manual pallet fork trolley



Fig. 4: Container with castor wheels

Figures 1 and 2 show two different methods of positioning the containers in a warehouse. For sites with a large fleet, there may be less floor space in the warehouse to have all the containers at floor level and may need to be stacked on shelves. Stacking introduces some issues with the requirements of using a fork lift truck every time when parts are needed to be placed in the containers on higher levels.

Figure 4 shows a container with castor wheels where it can be manually pushed in and out for stocking. This container also has the option of ventilating holes on the side which may be desired for specific purposes. Please note that the containers shown in figures 1, 2 and 3 do not have louvers at the top of the sides. They are molded corrugations for added strength and rigidity.

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Figure 5 shows how parts for multiple jobs can be organized in their own individual boxes contained within the container. This also aids in the workshop when multiple technicians are on the job and saves time trying to sort through parts for different jobs. Figure 6 shows the typical containers for smaller traditional machines. They are useful to the point when some larger parts are ordered. Such larger parts will require a separate location for storage.



Fig. 5: Enough room for 1 PM Service and 4 backlog jobs.



Fig. 6: Typical small containers used for smaller traditional machines.

3.0 Implementation Steps

- Review feasibility of using containers of suitable size.
- Research on such containers. A few suggestions to search for: pallet bins, parts containers, mega bins, warehouse containers.
- Set out area for the parts staging area.
- Develop procedure of backlog management process if not already done so. Refer to another Best Practice; Backlog Management 1007-2.0-1103 August 2006.
- Review availability of equipment to handle containers of desired size.
- Purchase containers and utilize.

4.0 Benefits

There are multiple benefits when using such containers on site.

- It will assist in contamination control efforts.
- Parts are collated in one container and have less tendencies to be misplaced. Misplaced parts cost time and labor to locate. In some cases, they must be reordered and it will increase downtime of the machine. If a reordered part does not have immediate delivery, the machine may have to return to work with a defect which may fail while in operation.

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- c) It promotes better control of the backlog jobs when the parts are all in one container.
- d) Moving the whole container out to the workshop as a carrier saves time in getting the parts to the machine in one single trip.
- e) Technicians get started on the jobs earlier.
- f) Lower machine downtime.
- g) Reduced frustrations of technicians and warehouse personnel.
- h) Professional environment promotes professional work.

5.0 Resources Required

- a) Set out appropriate space.
- b) Mark out space with lines.
- c) Handling equipment (manual lift truck, powered lift truck).

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

- Parts Management Best Practice
- Backlog Management Best Practice

8.0 Acknowledgements

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